

Trainings ▾

Preparatory Steps

Industrial and Power Generation

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative(-) battery cable first, and attach the negative (-) battery cable last.



⚠ WARNING ⚠

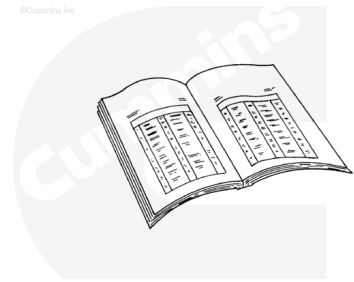
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Disconnect the batteries. See equipment manufacturer service information.
- Disconnect the air starter to prevent accidental engine starting. Refer to Procedure 012-022 in Section 12.
(/qs3/pubsys2/xml/en/procedures/56/56-012-022.html)
- Remove the drive unit and all related components. See equipment manufacturer service information.
- Remove the engine speed sensor. Use the following procedure in the QSK19, QSK23, QSK45, QSK60, and QSK78 Electronic Control System Troubleshooting and Repair Manual, Bulletin 3666113. Refer to Procedure 019-042 in Section 19.
(/qs3/pubsys2/xml/en/procedures/19/19-019-042.html)

QSK60 Marine Applications

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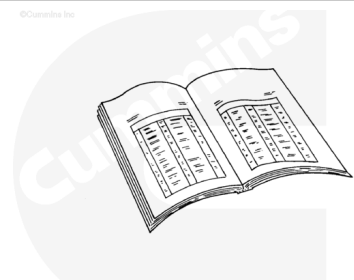
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(/qs3/pubsys2/xml/en/procedures/56/56-012-022.html)
- Remove the air cleaner assembly. Refer to Procedure 010-013 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-013-tr.html)
- Remove the air cleaner support bracket. Refer to Procedure 016-006 in Section 16.
(/qs3/pubsys2/xml/en/procedures/56/56-016-006-tr.html)
- Remove the drive unit and all related components. See equipment manufacturer service information.

with Electronically Actuated Injector

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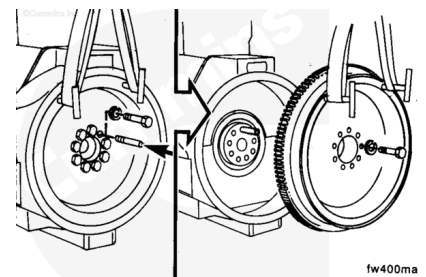
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- Disconnect the air starter to prevent accidental engine starting. Refer to Procedure 012-022 in Section 12. (</qs3/pubsys2/xml/en/procedures/56/56-012-022.html>)
- Remove the drive unit and all related components. See equipment manufacturer service information.
- Remove the engine speed sensor. Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System) Troubleshooting and Repair Manual, Bulletin 4021533. Refer to Procedure 019-042 in Section 19. (</qs3/pubsys2/xml/en/procedures/05/05-019-042.html>)

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : For the QSK45 excavator applications that use the 12-point head capscrews, remove and discard the flywheel, capscrews, and spacer at end of every life rebuild.

Remove two capscrews and install two guide studs.

Install two T-handles.

Install the lifting straps.

Remove the remaining capscrews.

A mallet can be used to tap the flywheel, if necessary.

Remove the flywheel.

Remove and discard the roll pin in the rear face of the crankshaft.

Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

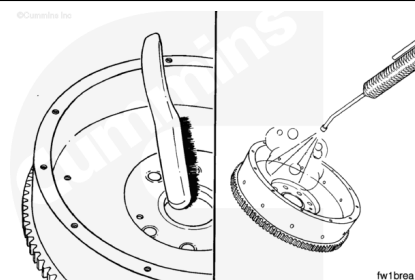
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent or steam to clean the flywheel.

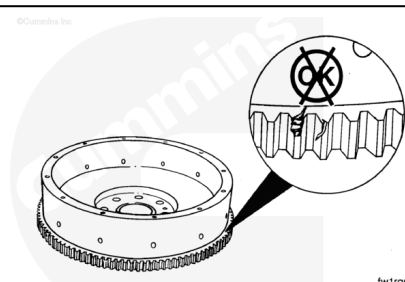
Use a wire brush to clean the crankshaft pilot bore.

Dry with compressed air.



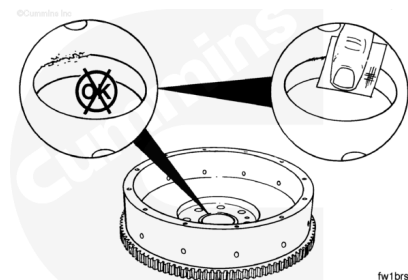
Inspect the teeth of the ring gear for wear or damage.

If the teeth of the ring gear are worn or damaged, the ring gear **must** be replaced. Refer to Procedure 016-008 in Section 16. (</qs3/pubsys2/xml/en/procedures/56/56-016-008-tr.html>)



Use Cummins® dye penetrant kit, Part Number 3375432, to check the area of the mounting capscrew holes for cracks. There **must not** be any cracks in the flywheel mounting area.

Inspect the pilot bearing mounting hole for damage. If the pilot hole is damaged, the flywheel **must** be replaced.

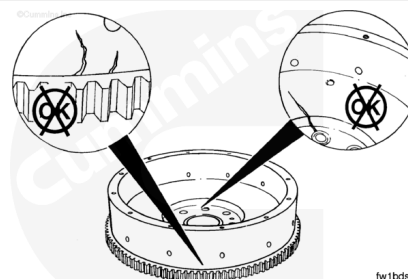


⚠ WARNING ⚠

Do not use a cracked flywheel. A cracked flywheel can break and cause personal injury or property damage.

Inspect the face of the flywheel for cracks or other damage.

The flywheel **must** be replaced if cracked or otherwise damaged.



Install

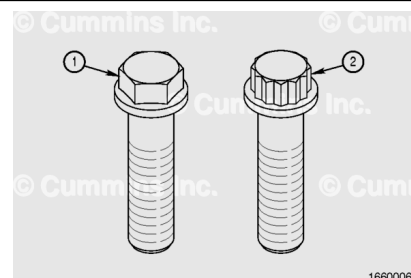
Note : There are two types of flywheel mounting capscrews used on the QSK45 and QSK60 engines.

The capscrews can be identified by having either a 6-point head or a 12-point head. Engines using the 12-point head capscrews also have a clamping ring installed.

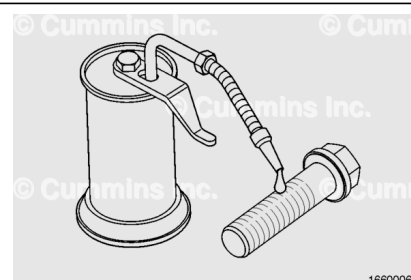
1. 6-point head capscrew
2. 12-point head capscrew.

Note : The 12-point head capscrew is **only** used on the QSK45 engine in certain excavator applications.

Note : For the QSK45 excavator applications that use the 12-point head capscrews (2), remove and discard the flywheel, capscrews, and spacer at end of every life rebuild.

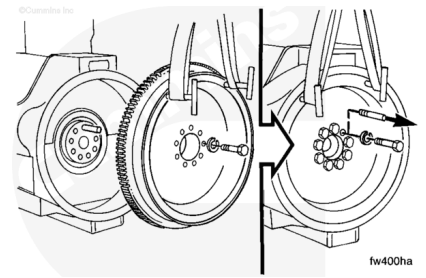


Lubricate the flywheel mounting capscrew threads and flange head surfaces with clean engine oil.



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⚠ WARNING ⚠

Install a new roll pin into the dowel hole in the rear face of the crankshaft to achieve the correct protrusion, 8.7mm +/- 1.0mm.

Install two guide studs to prevent the flywheel from rotating.

Install the flywheel and line up the dowel hole in the flywheel with the roll pin in the crankshaft.

Install the flywheel mounting capscrews.

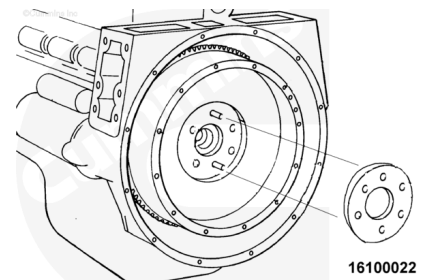
Remove the two guide studs and install the last two flywheel mounting capscrews.

Remove the two T-handles.

Note : Perform this step only for QSK45 applications that use the 12-point head capscrews and clamping ring.

Insert the clamping ring against the flywheel. The clamping ring can be installed with either side toward the flywheel.

Install the 12-point head flywheel mounting capscrews.



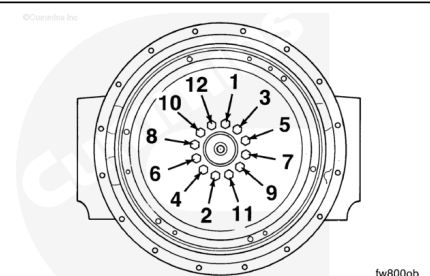
For the 6-point head capscrews:

Tighten the flywheel mounting capscrews in the sequence shown.

Torque Value:

1. 200 n•m [148 ft-lb]
2. 380 n•m [280 ft-lb]
3. 685 n•m [505 ft-lb]

For the 12-point head capscrews:



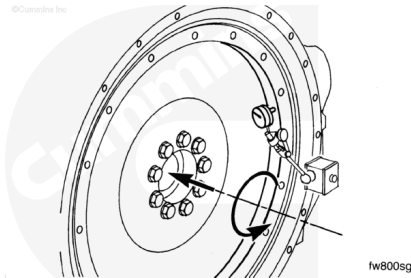
Tighten the flywheel mounting capscrews in the sequence shown.

Torque Value:

- 1. 200 n•m [148 ft-lb]
- 2. 380 n•m [280 ft-lb]
- 3. 685 n•m [505 ft-lb]
- 4. 800 n•m [590 ft-lb]

Measure the flywheel total indicator runout.

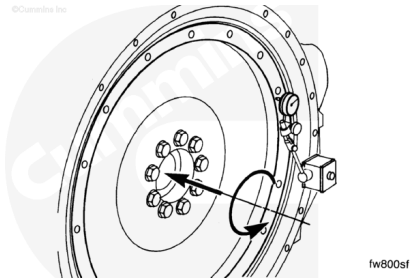
Measurements				
		mm		in
Flywheel Housing	Number 0	648		25.5
	Number 00	787		31.0
Flywheel Bore Total Indicator Runout				
	mm		in	
Number 0		0.127	MAX	0.005
Number 00		0.127	MAX	0.005



The maximum allowable flywheel face runout is determined by the distance from the center of the flywheel to the contact point of measurement, divided by 1000.

Measure the flywheel face total runout.

Flywheel Face Total Indicator Runout				
	mm		in	
Number 0 (with a 254 mm [10 in] center distance)		0.254	MAX	0.010



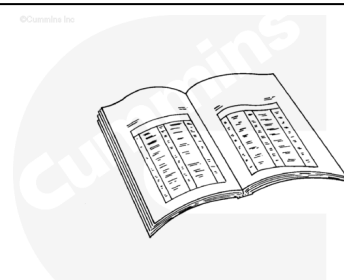
Number 00 (with a 305 mm [12 in] center distance)		0.305	MAX	0.012
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Finishing Steps

Industrial and Power Generation

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- Connect the batteries. See equipment manufacturer service information.
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QSK60 Marine Applications

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